When do GPT-2's attention heads rest? Three results and a proposal

Pablo Leyva · meeting prep · GPT-2 small, BOS + 32 tokens, attention measured at the final token · **resting** = late-layer (6–11) attention parked on the first token (BOS), the candidate no-op · all predictions pre-registered

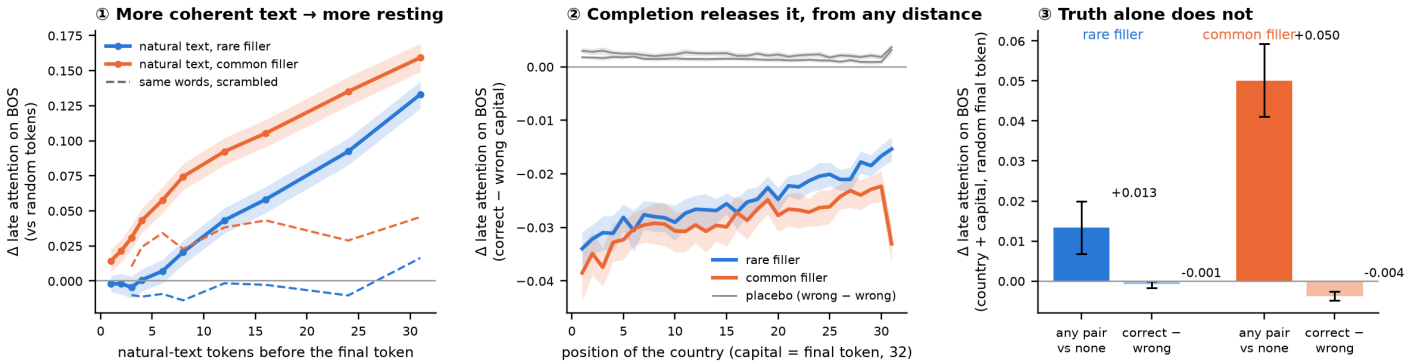
Summary

① **More coherent text → more resting.** Late heads' BOS attention rises steadily with the amount of natural text before an unrelated final token (+0.13 / +0.16 at 31 tokens, vs random tokens), needs word order, and keeps 65–72% when the last two words are randomized.

② **Completion releases it.** When the final token completes the context (a country's capital, a phrase's object, a passage's next word), late heads rest less (-0.02 to -0.06); for country→capital this held from every one of the 31 positions.

③ **Truth alone does not.** A correct vs wrong country–capital pair before an unrelated final token barely moves resting (-0.001 / -0.004), though GPT-2 knows all 11 capitals; any real pair raises it (+0.013 / +0.05).

⇒ **Working hypothesis:** heads rest when there is coherent context the current token does not use ("nothing to integrate").



Blue / orange = rare / common random filler tokens (two independent replications). Bands and bars = 95% bootstrap CIs over 50–200 rows.

① **How (E3.2).** The last k = 1–31 tokens of 12 hand-written passages before a random final token; 200 rows $\times$ 2 filler types. Controls: the same tokens scrambled, and same-form random words. 12/12 passages positive from k = 12. Overturns our earlier "last word pair" account, which held only for 3-word phrases.

② **How (R1-in + sweep).** Every country paired with every capital, so each capital is the final token once correctly and ten times wrongly: token identity cancels in correct − wrong. Adjacent: -0.018 / -0.034, 11/11 countries, permutation $p < 10^{-4}$. Same direction for cities, phrase objects (-0.03 to -0.05) and natural next words (-0.06, 12/12 passages).

③ **How (R1-out).** Same grid with the pair at 30–31 and a random final token. Manipulation check: at the country, GPT-2 prefers its own capital in 11/11 countries (+2.4 / +2.7 nats). The pairing shows up elsewhere: correct pairs shift attention inside the pair from the capital to the country.

Why trust it: TransformerLens and Hugging Face agree to $< 5 \times 10^{-5}$; reruns are bit-identical; placebo contrasts flag $\leq$1 head (0 of 496 sweep placements); a positive control works (known induction heads L5H5, L6H9 switch from BOS to the copy target); 19 of 37 pre-registered predictions held, and the failures are reported. **Limits:** GPT-2 small only; random-token contexts except E3.2; some effects differ between the two filler types.

Proposed next direction: is resting actually a no-op?

Why this, causally. ①–③ show that BOS attention tracks whether the current token has context to integrate, but every result so far is an *attention weight*; "no-op" is still an interpretation. The results also hand us the tool to test it: ① and ② are two dials that move resting in opposite directions while holding the tokens fixed (amount of coherent text; correct vs wrong completion with the same final token). So we can change resting without changing the input tokens, and ask whether the computation changes with it.

Test. (1) *Value-weighted attention:* does a resting head's output ($\sum_k \alpha_k v_k$) shrink toward zero as its BOS weight rises? (2) *Ablation:* mean-ablate each late head and measure the change in the model's next-token distribution (KL) in high-resting vs released contexts from the same grids.

Prediction and stakes. A no-op predicts near-zero output and near-zero ablation cost while heads rest, and a larger cost once completion releases them. Confirmed, "resting = no-op" becomes a measured property and ②'s completion signal becomes the target for circuit work (patching what tells late heads to stop resting). Refuted, BOS attention does real work and the thesis has to change.